

Deep Research Prompt: Design a Self-Improving AI Sales Agent with Persistent Memory

You are a world-class AI systems architect, ex-OpenAI, ex-DeepMind, and ex-Palantir product designer.

Your task is NOT to build another meeting summarizer or chatbot.

Your objective is to invent a next-generation **Self-Improving AI Sales Agent** whose biggest competitive advantage is **persistent long-term memory**.

The AI should become smarter after every interaction and eventually act like the company's best sales engineer.

Background

Current AI assistants are stateless.

Every meeting starts from scratch.

Even if transcripts are saved, they are simply searchable documents.

The AI never develops an evolving understanding of the customer.

Your mission is to design an AI system that continuously builds a living memory of every prospect and organization.

Think of it as creating a "Customer Digital Twin."

Core Vision

Instead of remembering conversations...

The AI should remember knowledge.

It should continuously update:

- customer profile
- company profile
- buying stage
- objections
- competitors
- technical stack
- stakeholders
- previous promises

- decision history
- budget signals
- urgency
- sentiment
- negotiation style
- product interests
- unresolved questions
- expansion opportunities
- churn risks

The AI should maintain these memories for months or years.

Self-Improvement Requirement

After every interaction the system should automatically:

1. Analyze the conversation.
2. Decide what new information was learned.
3. Detect contradictions with previous memory.
4. Update confidence scores.
5. Merge duplicate entities.
6. Detect new buying signals.
7. Detect new risks.
8. Generate new hypotheses.
9. Update customer strategy.
10. Improve future recommendations.

The AI must become smarter after every meeting.

Memory Architecture

Design a complete long-term memory architecture.

Include:

Episodic Memory

Store every meeting as an event.

Example

Meeting

↓

People

↓

Topics

↓

Decisions

↓

Emotions

↓

Promises

↓

Questions

↓

Outcome

Semantic Memory

Extract stable knowledge.

Example

Company uses Salesforce.

Decision maker is CTO.

SOC2 required.

Budget approved.

Current vendor is HubSpot.

Procedural Memory

Remember successful sales strategies.

Example

Healthcare companies respond well to ROI case studies.

Security-first customers require compliance documentation early.

Finance teams negotiate hardest at quarter end.

Relationship Memory

Maintain connections between

People

Departments

Companies

Competitors

Products

Previous deals

Support tickets

Emails

Meetings

Reflection Memory

After every meeting ask

"What should I learn from this?"

Store lessons.

Example

Last pricing discussion failed because customer still doubted scalability.

Next meeting should demonstrate benchmark results.

Multi-Agent Design

Design specialized AI agents.

Examples

Conversation Agent

Memory Agent

CRM Agent

Research Agent

Strategy Agent

Proposal Agent

Negotiation Agent

Reflection Agent

Learning Agent

Manager Agent

Explain how they communicate through shared memory.

Continuous Learning Loop

Design a system that continuously improves.

Meeting

↓

Transcript

↓

Knowledge Extraction

↓

Memory Update

↓

Reflection

↓

Strategy Update

↓

CRM Update

↓

Email Draft

↓

Follow-up Tasks

↓

Preparation for Next Meeting

↓

Repeat forever

Proactive Intelligence

The AI should never wait for user prompts.

Examples

Notify salesperson:

"The customer mentioned SOC2 three meetings ago but you still haven't sent documentation."

"The CTO has attended three meetings while procurement has never joined."

"The buying committee expanded."

"Customer sentiment has decreased."

"This account resembles three previous successful enterprise deals."

Predictive Intelligence

Instead of reporting the past...

Predict the future.

Estimate

Probability of closing

Risk of churn

Likelihood of competitor win

Budget approval probability

Decision timeline

Next likely objection

Best follow-up action

Best stakeholder to involve

Integrations

Assume integrations with

Salesforce

HubSpot

Slack

Zoom

Google Meet

Microsoft Teams

Gmail

Outlook

LinkedIn

Notion

Jira

GitHub

Zendesk

Intercom

Calendar

Email

CRM

Support systems

Marketing automation

Competitive Analysis

Compare against

Gong

Clari

Fireflies

Avoma

Fathom

Salesforce Einstein

Microsoft Copilot

Google Gemini Workspace

OpenAI ChatGPT

Explain why persistent memory creates a defensible moat.

Hackathon Deliverables

Design a demo that can be shown in under 5 minutes.

Include

Architecture

Memory Model

Agent Workflow

UI Mockups

Data Flow

Technology Stack

Vector Database

Knowledge Graph

Memory Schema

Prompt Design

Reflection Loop

Evaluation Metrics

Demo Story

Judge Pitch

Stretch Goal

Imagine this product becomes the world's best AI Sales Engineer in five years.

Describe

its architecture,

business model,

technical moat,

network effects,

and why it becomes difficult for competitors to copy.

Think from first principles.

Do not optimize for simplicity.

Optimize for creating a category-defining AI company built around long-term memory rather than chat history.